```
unaids.remainder, step 2 (2026-11-02)

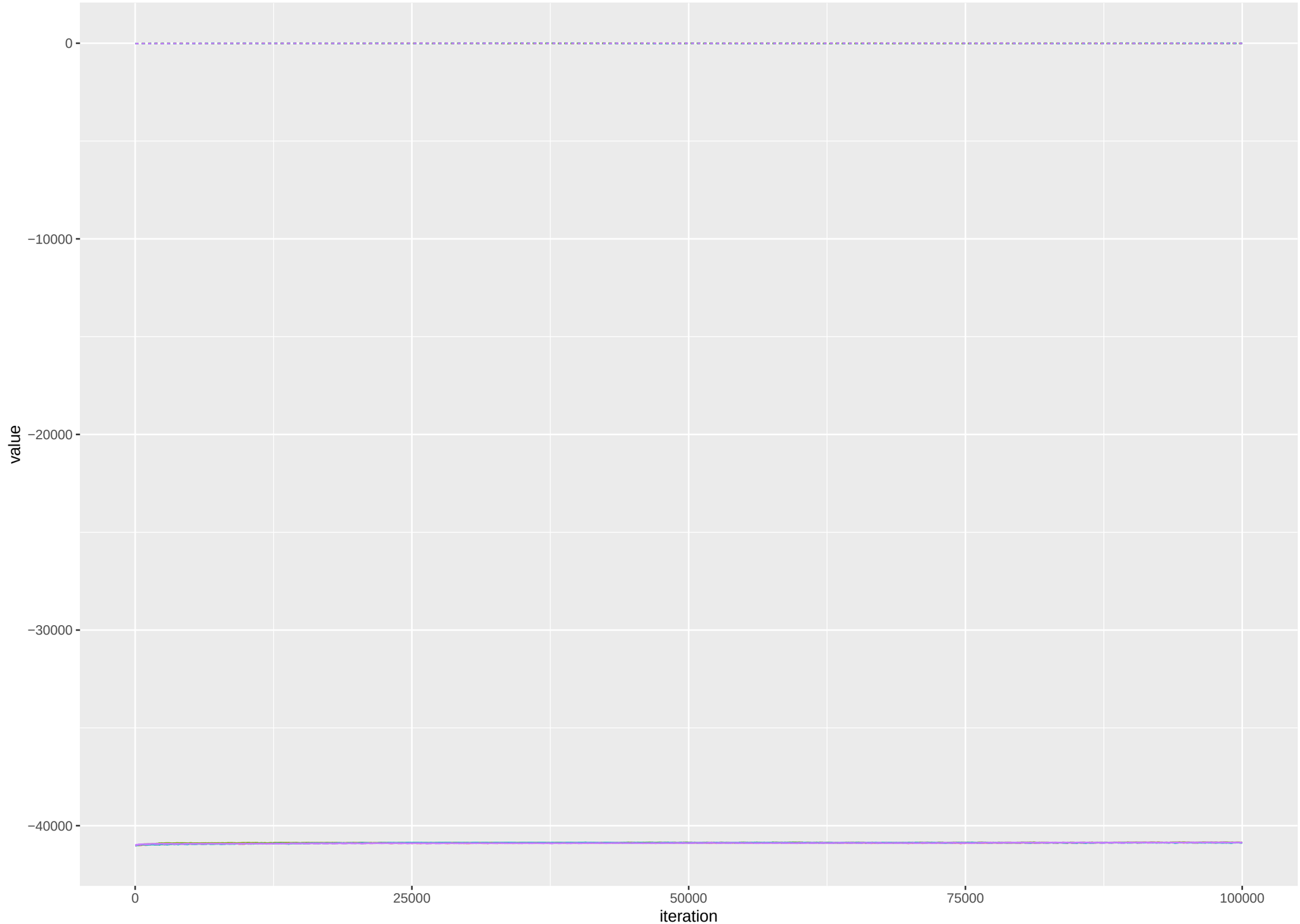
 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277                -40873.9           -40851.7                 0
     2 20263017             5000           0.277                -40850.7           -40842.3                 0
     3 20263018             5000           0.275                -40871.1           -40857.0                 0
     4 20263019             5000           0.275                -40860.9           -40849.9                 0
                                                 file
 mcmc_unaids.remainder_step2_chain1_2026-10-30.Rdata
 mcmc_unaids.remainder_step2_chain2_2026-10-31.Rdata
 mcmc_unaids.remainder_step2_chain3_2026-10-31.Rdata
 mcmc_unaids.remainder_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
          age.20.to.29.transmission.multiplier.3                                        trate.3
                                           2.331                                          2.221
          age.15.to.19.transmission.multiplier.3       age.50.and.over.transmission.multiplier.3
                                           2.068                                          2.053
                                         trate.4         age.30.to.39.transmission.multiplier.3
                                           1.882                                          1.868
          age.20.to.29.transmission.multiplier.0         age.20.to.29.transmission.multiplier.1
                                           1.828                                          1.775
          age.15.to.19.transmission.multiplier.0                            age.assortativity
                                           1.731                                          1.690
                                         trate.1        age.25.to.49.hiv.mortality.multiplier.0
                                           1.685                                          1.678
  age.65.to.79.mortality.slope.multiplier.female                                        trate.0
                                           1.641                                          1.617
                male.hiv.mortality.multiplier.0                       female.to.male.multiplier
                                           1.563                                          1.485
          age.30.to.39.transmission.multiplier.0 age.45.to.64.mortality.intercept.multiplier.male
                                           1.482                                          1.345
                  hiv.specific.mortality.rates.0    age.45.to.64.mortality.slope.multiplier.male
                                           1.338                                          1.323
          age.0.to.14.hiv.mortality.multiplier.1   over.80.mortality.intercept.multiplier.female
                                           1.315                                          1.271
          age.20.to.29.transmission.multiplier.2                                        trate.2
                                           1.271                                          1.264
age.65.to.79.mortality.intercept.multiplier.male
                                           1.256

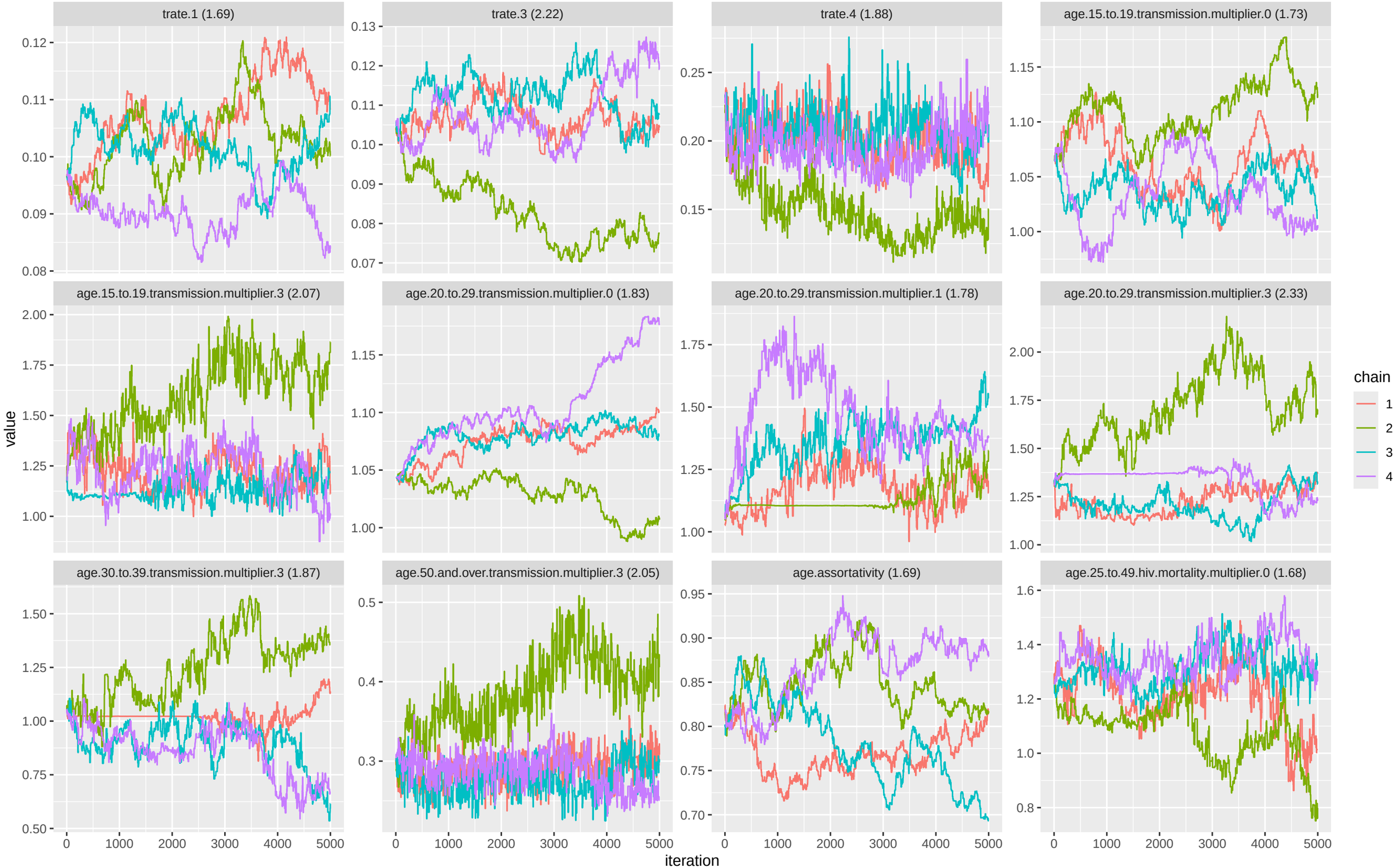
Merged simset:
         location simulations total.weight                                                 file
 unaids.remainder        1000         1000 simset_unaids.remainder_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 13 of 33 targets flagged
     data.type         age  sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality 50 and over both              26                   8              x2.88      2025            x2.86
 hiv.mortality       25-49 both              26                   0              x0.50      2025            x0.61
     awareness        0-14 both              11                   0          -13.6 pts      2025        -19.0 pts
 hiv.mortality       10-19 both              26                  15              x1.38      2025            x1.70
 hiv.mortality       15-24 both              26                  42              x1.44      2025            x1.74
    engagement        0-14 both              11                   9           -9.3 pts      2025         +2.1 pts
 hiv.mortality        0-14 both              26                  23              x1.22      2025            x1.17
    prevalence       15-24 both              26                  23              x1.10      2025            x1.14
   suppression        0-14 both              11                  27           -4.4 pts      2025         +5.2 pts
```

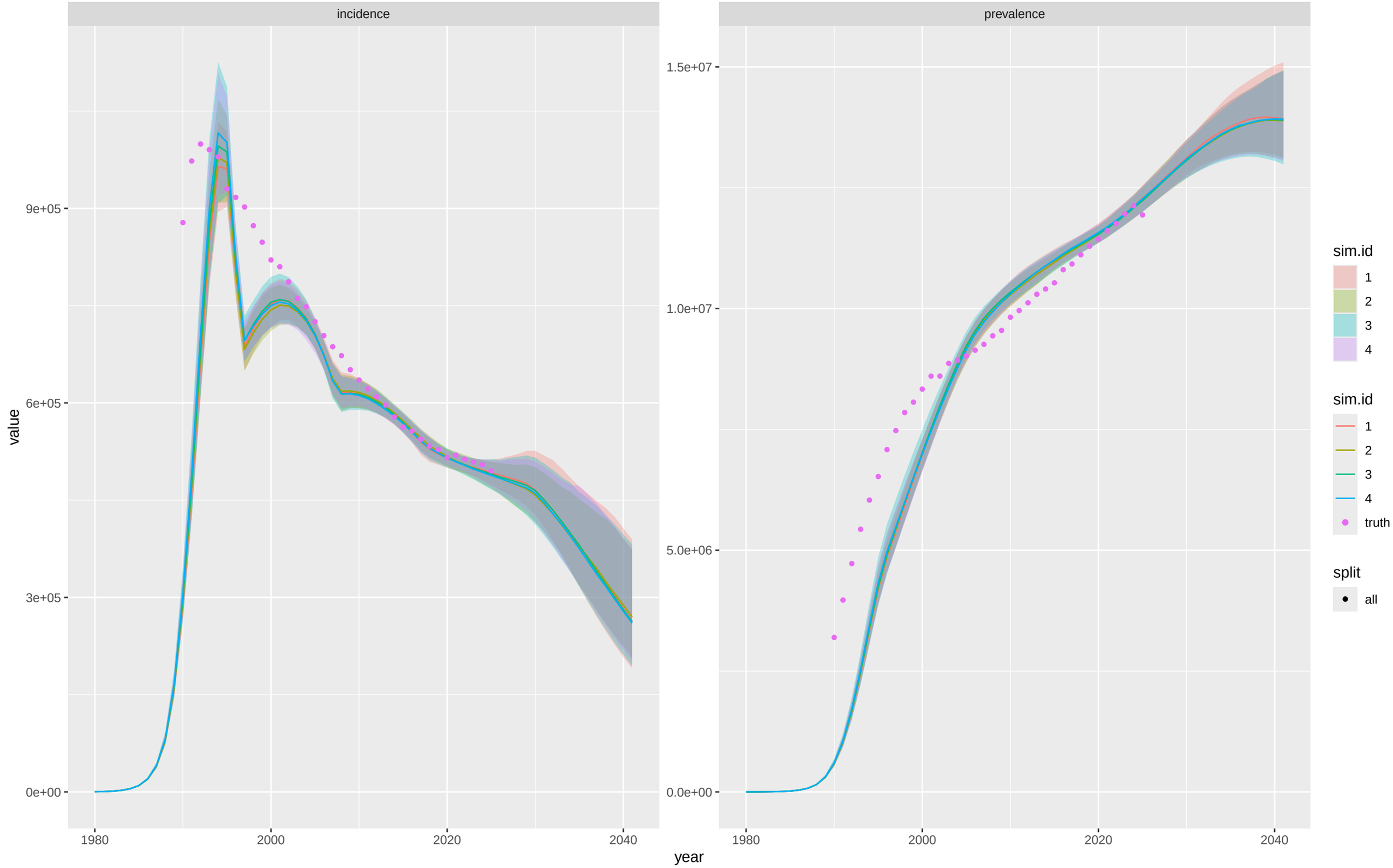
unaids.remainder step 2: likelihood



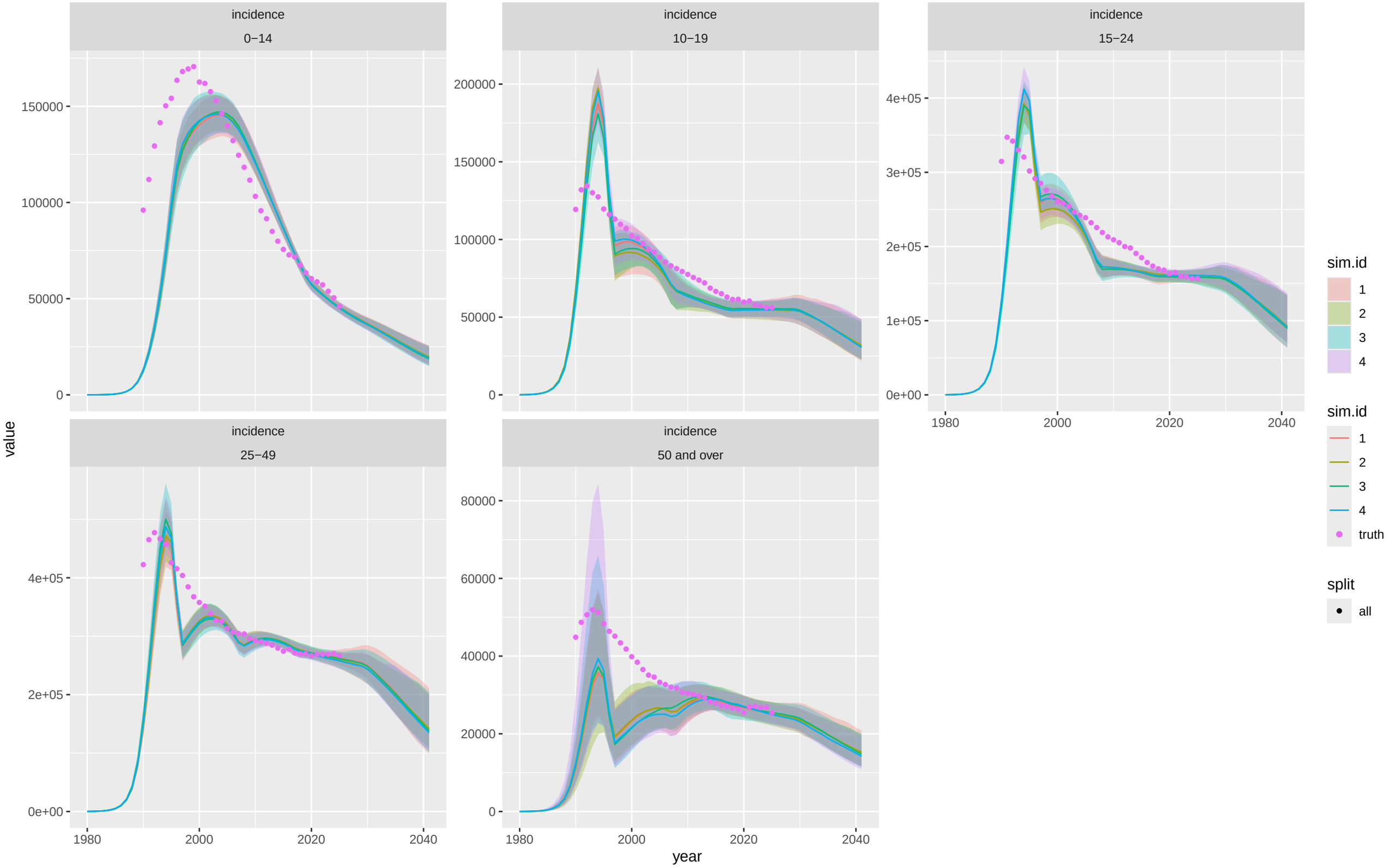
unaids.remainder step 2: trace



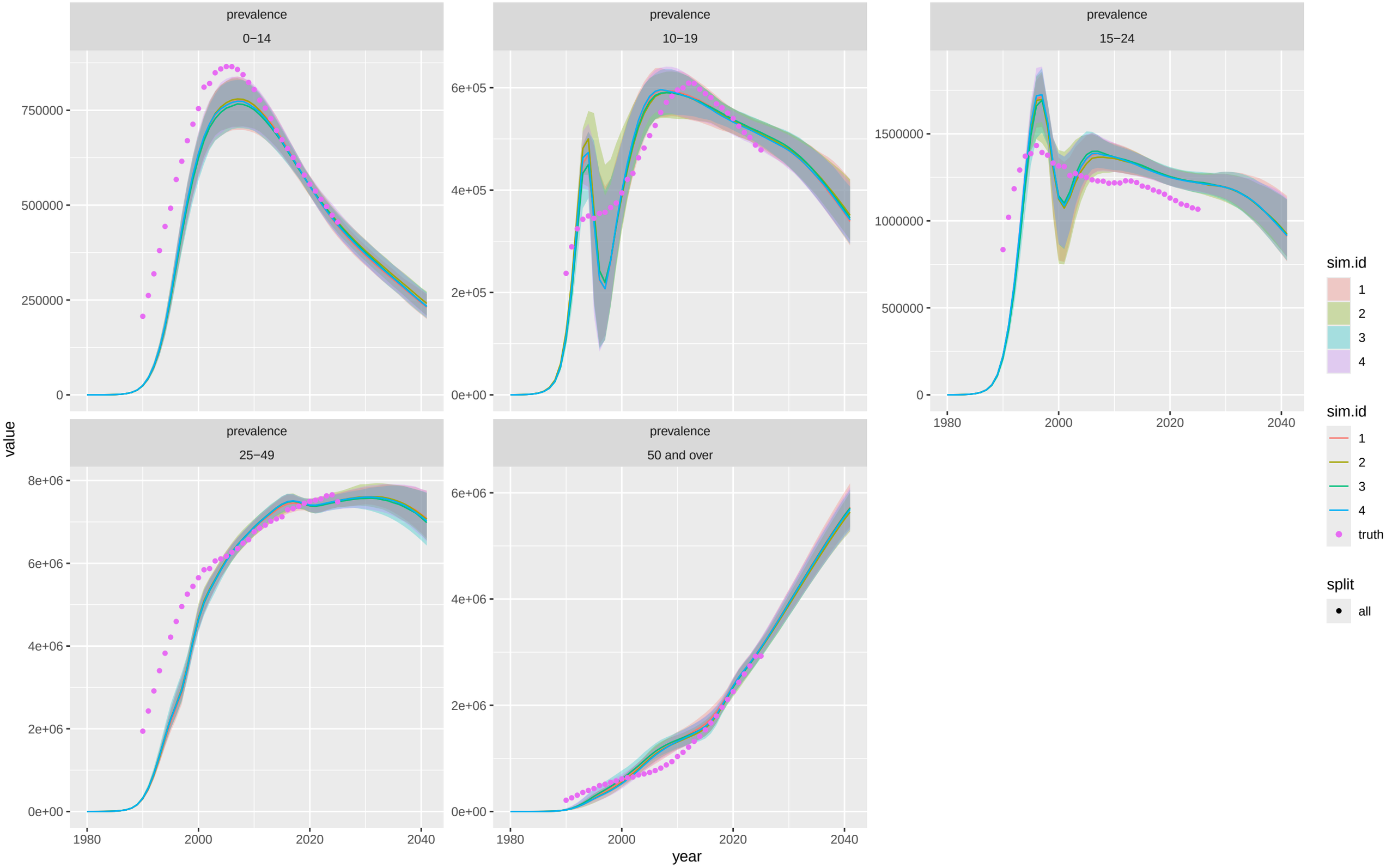
unaids.remainder step 2: fit (one color per chain)



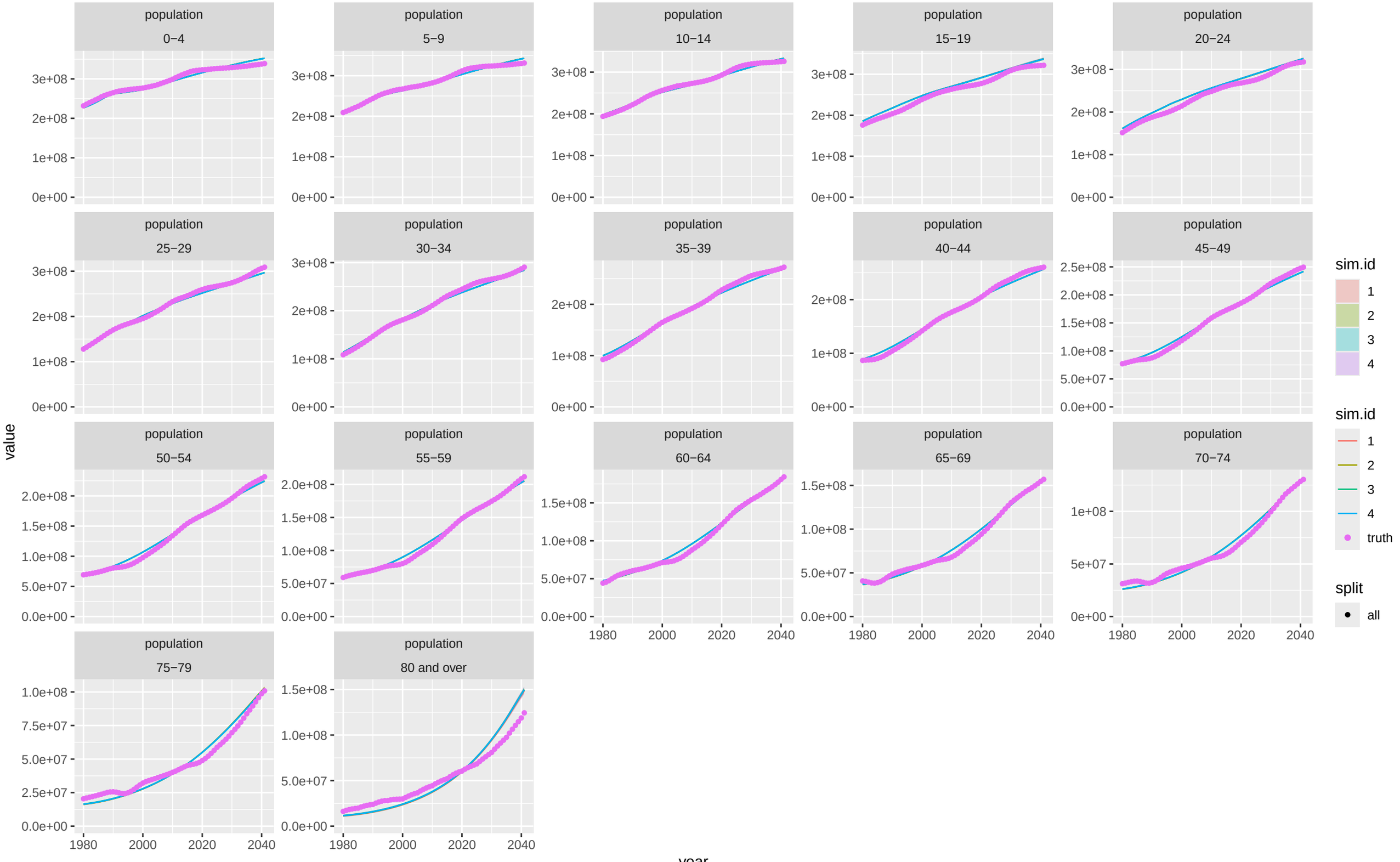
unaids.remainder step 2: incidence.by.age (one color per chain)



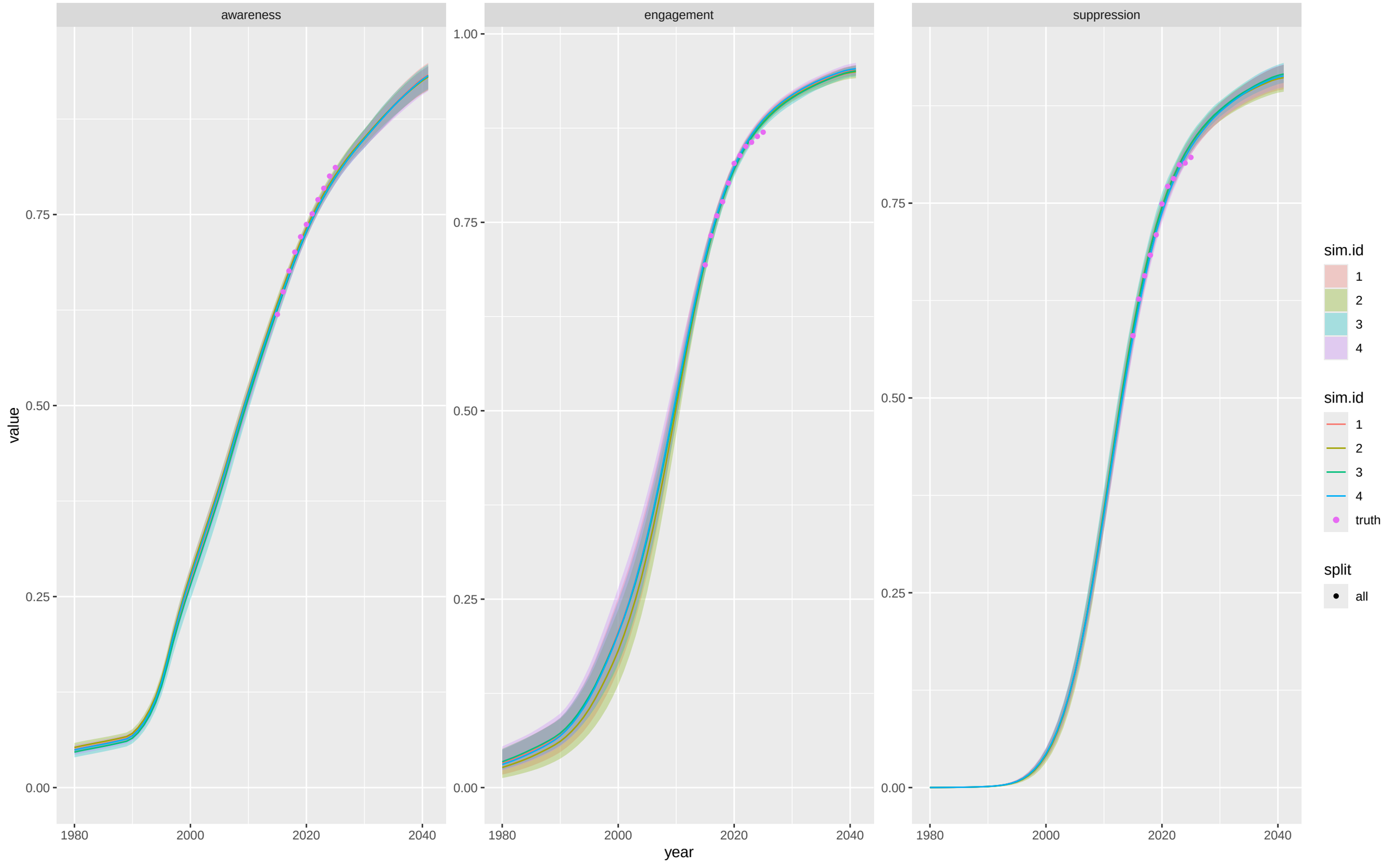
unaids.remainder step 2: prevalence.by.age (one color per chain)



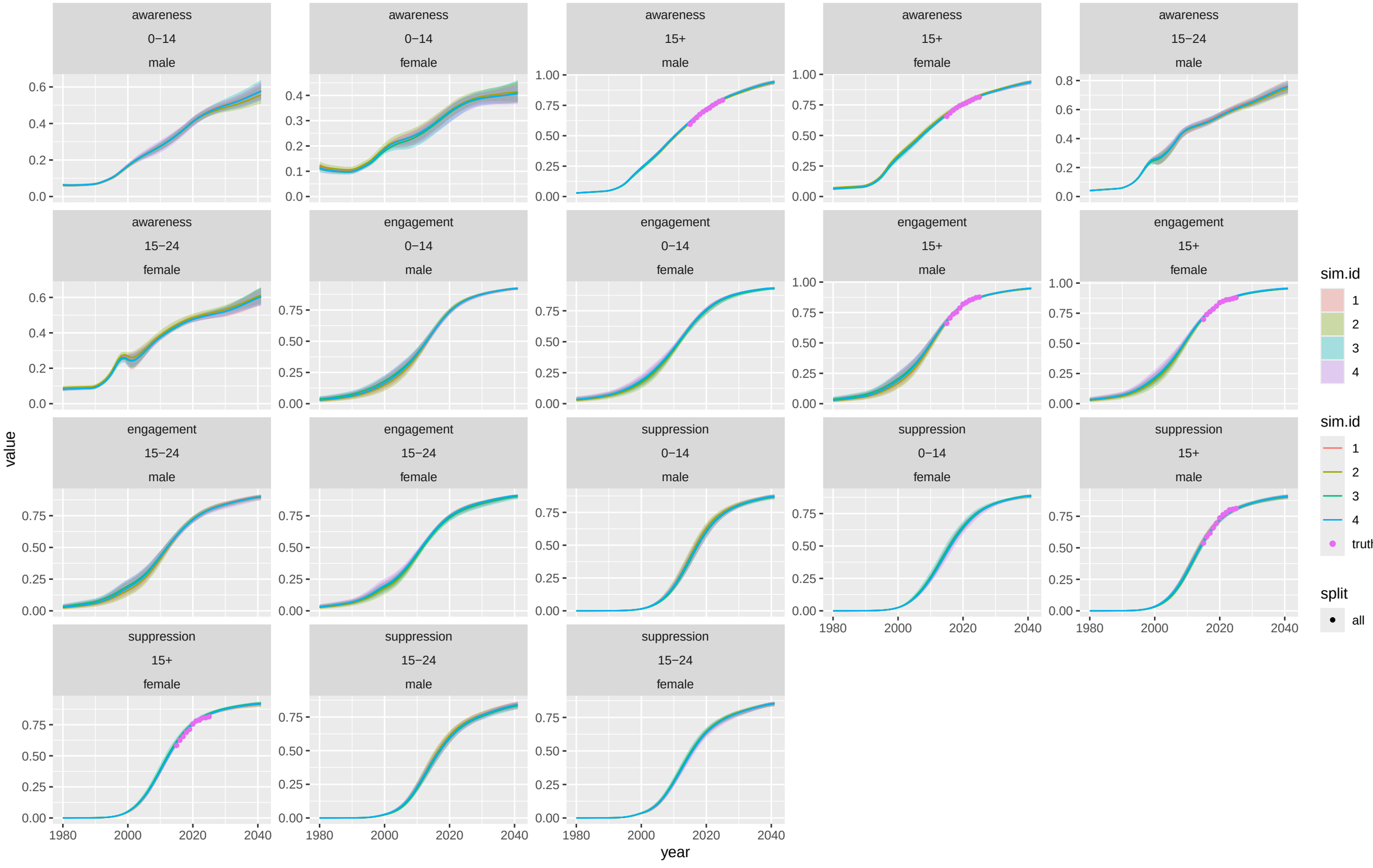
unaids.remainder step 2: population.by.age (one color per chain)



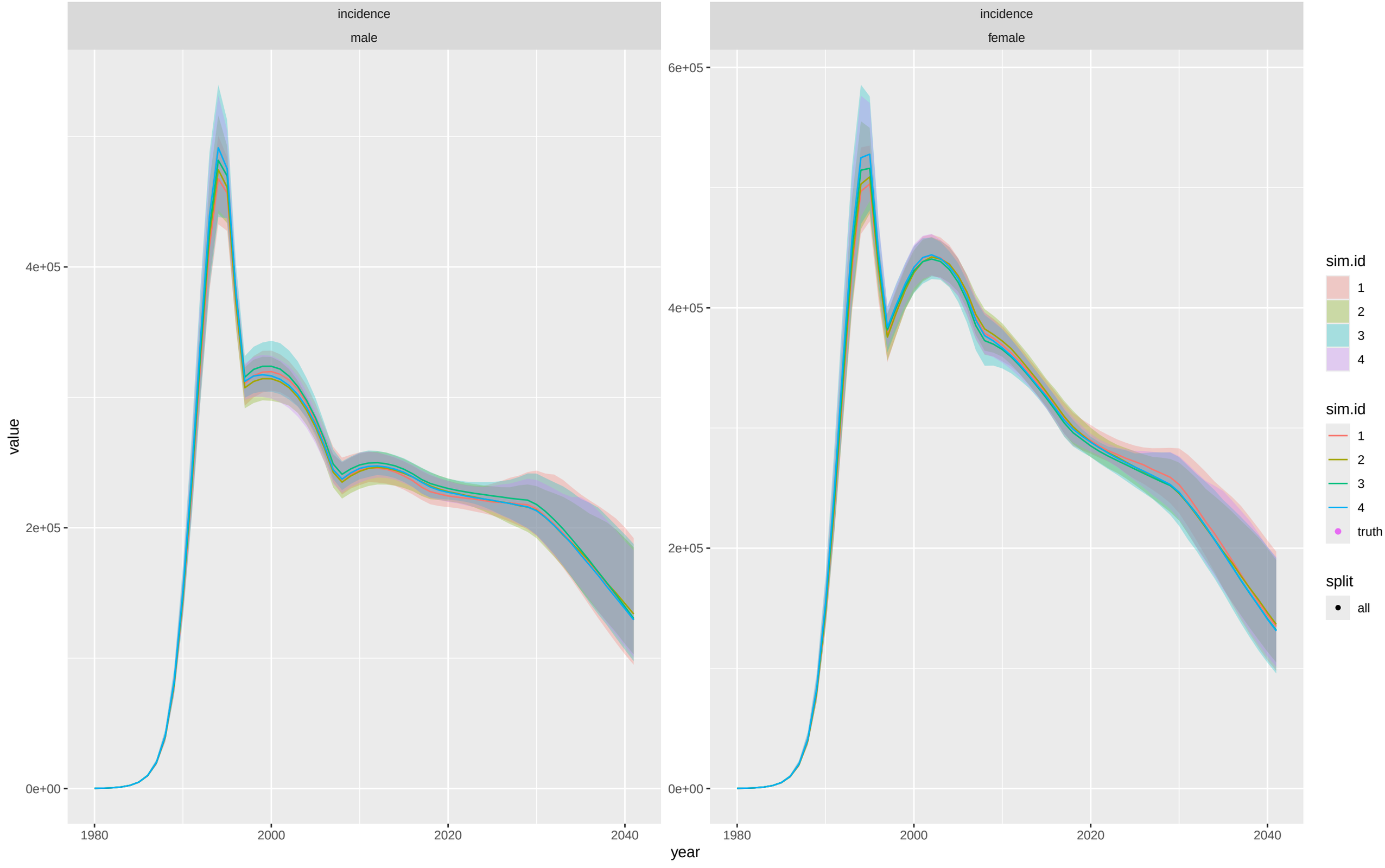
unaids.remainder step 2: cascade (one color per chain)



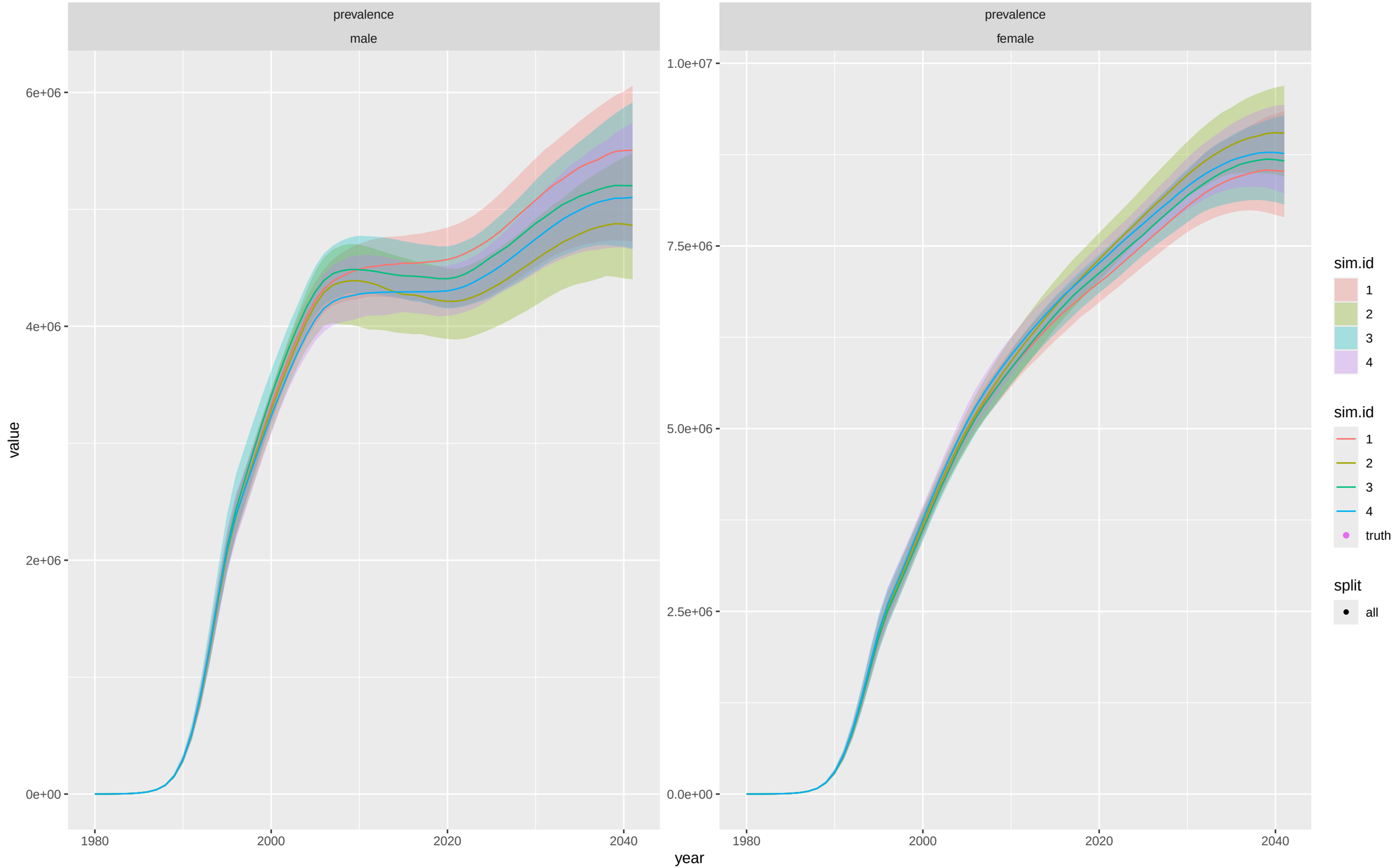
unaids.remainder step 2: cascade.by.age.sex (one color per chain)



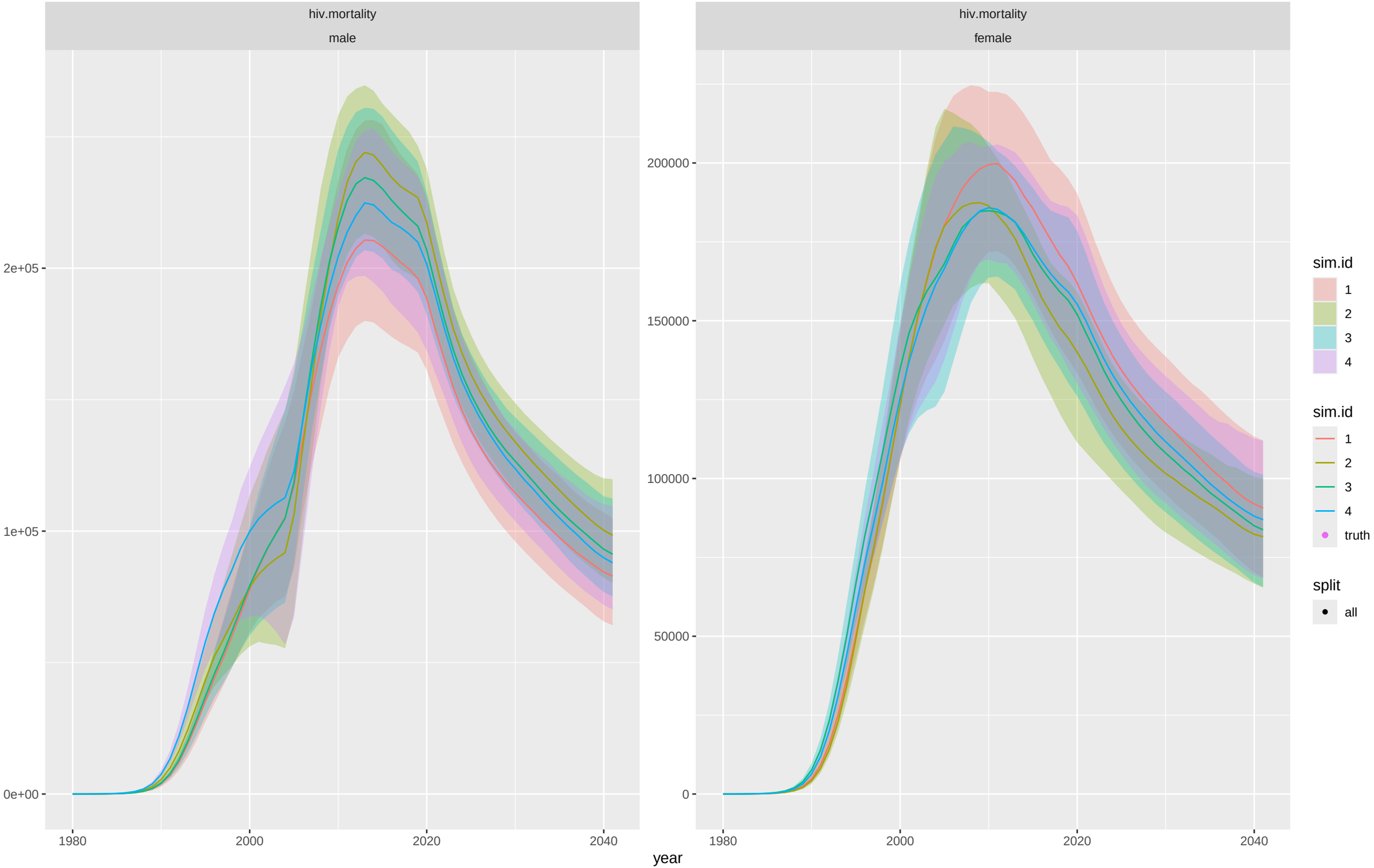
unaids.remainder step 2: incidence.by.sex (one color per chain)



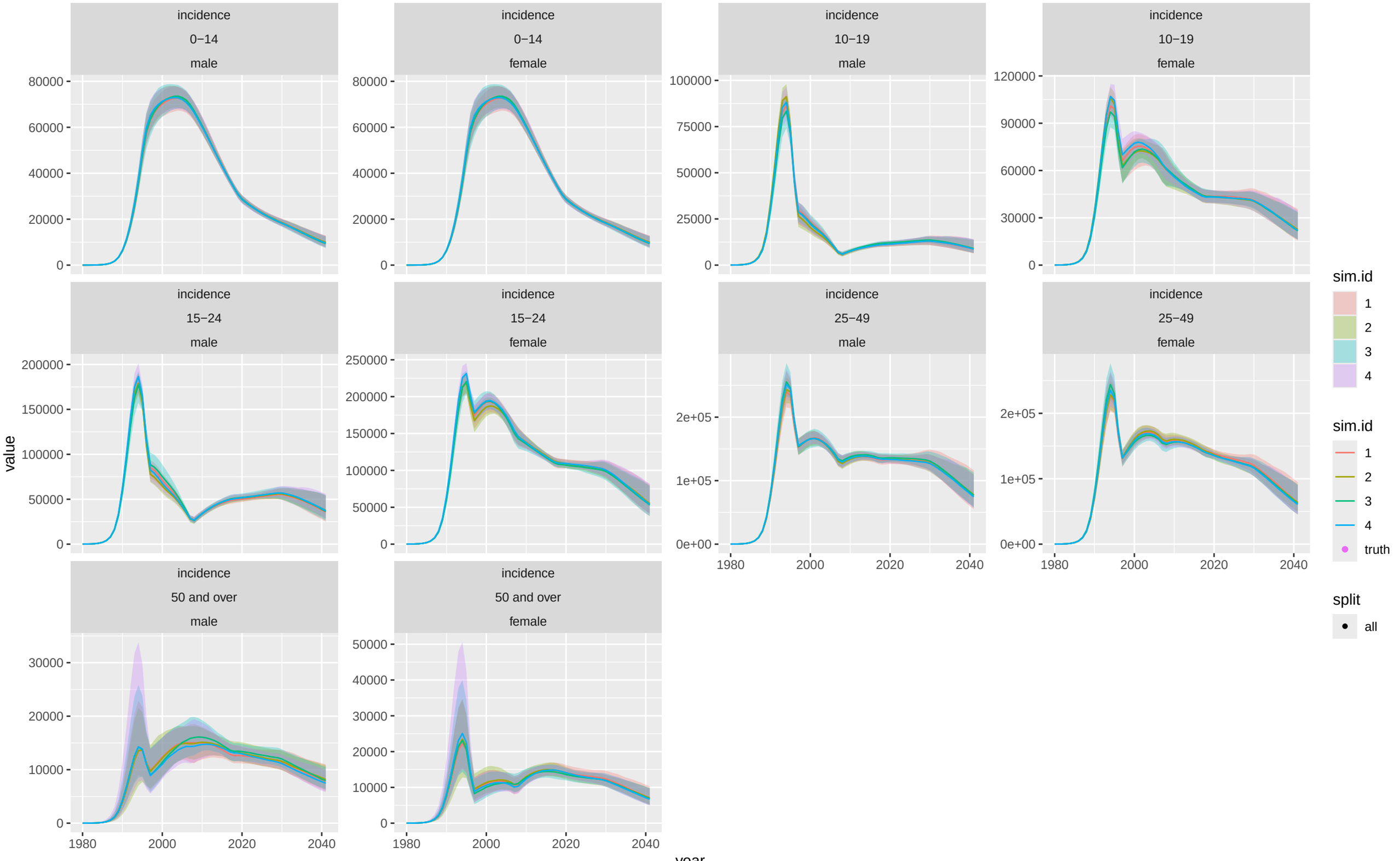
unaids.remainder step 2: prevalence.by.sex (one color per chain)



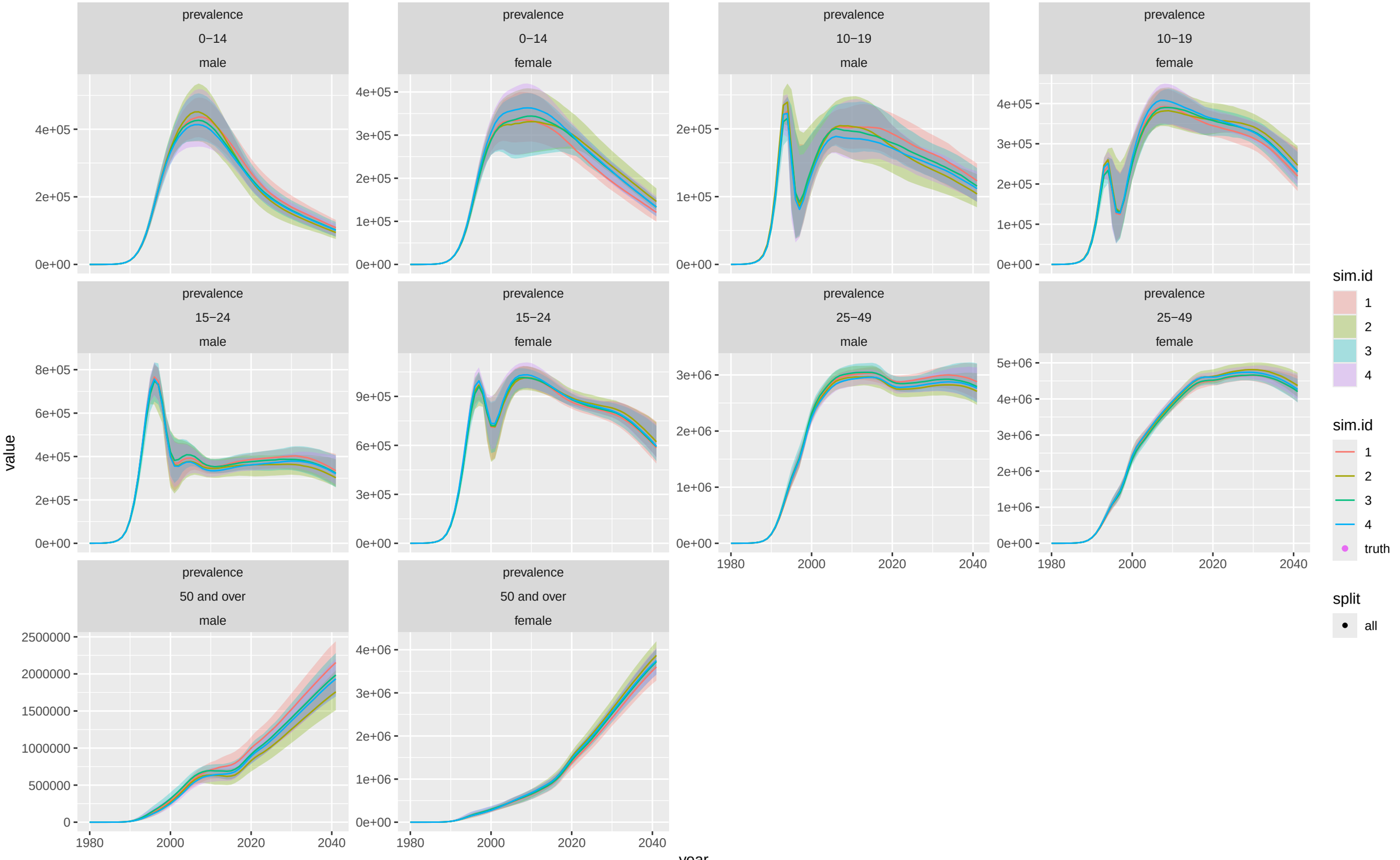
unaids.remainder step 2: hiv.mortality.by.sex (one color per chain)



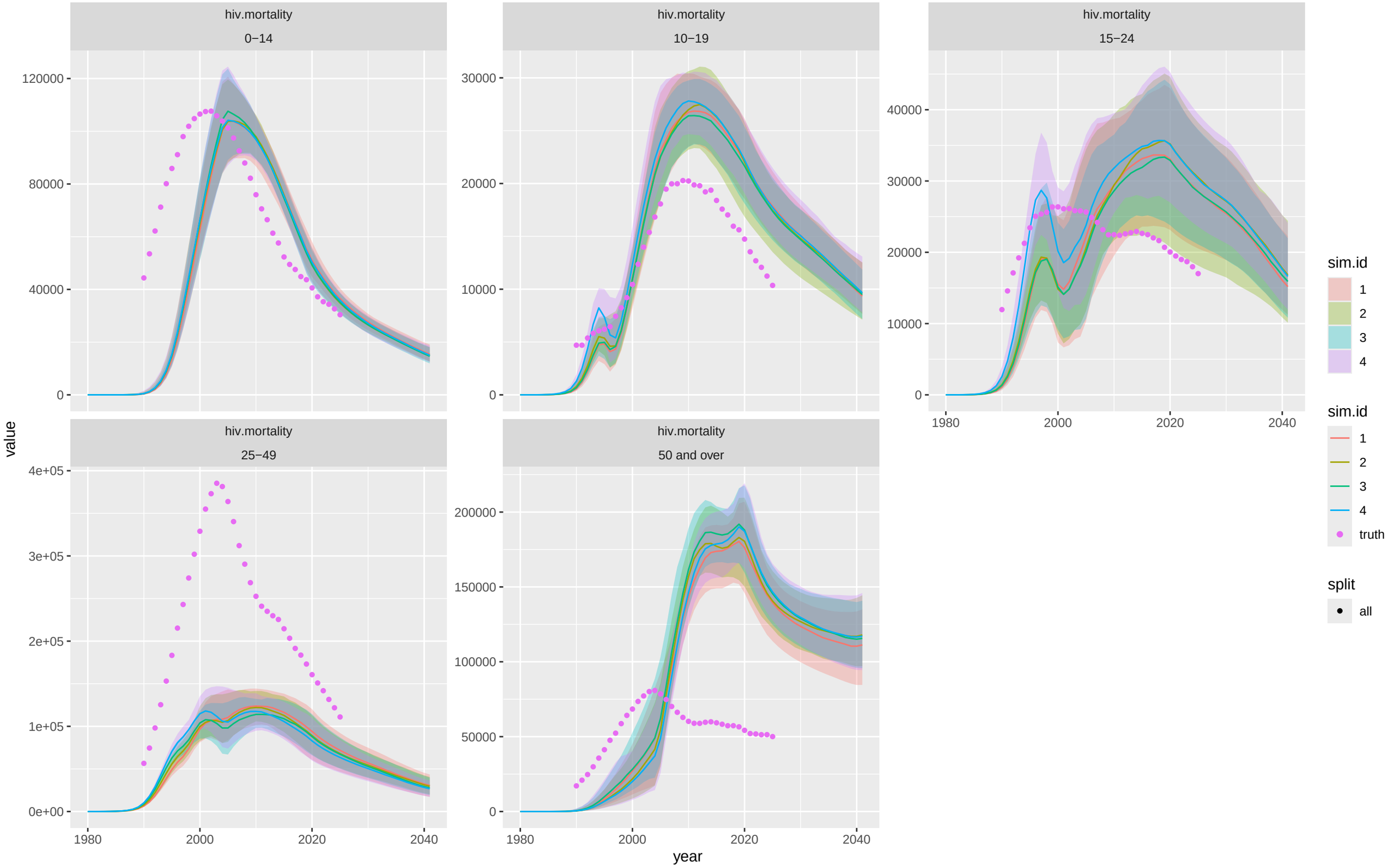
unaids.remainder step 2: incidence.by.age.sex (one color per chain)



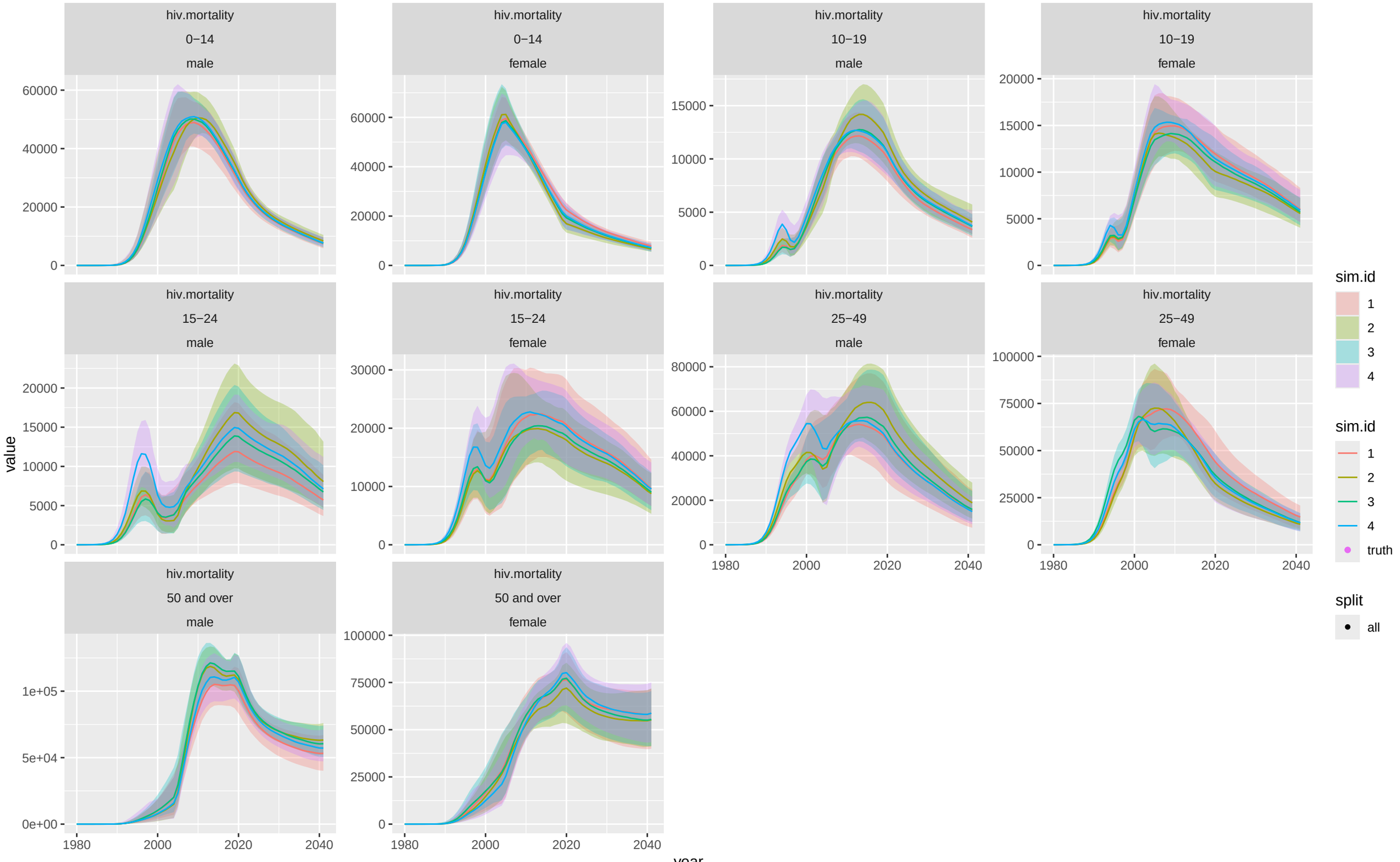
unaids.remainder step 2: prevalence.by.age.sex (one color per chain)



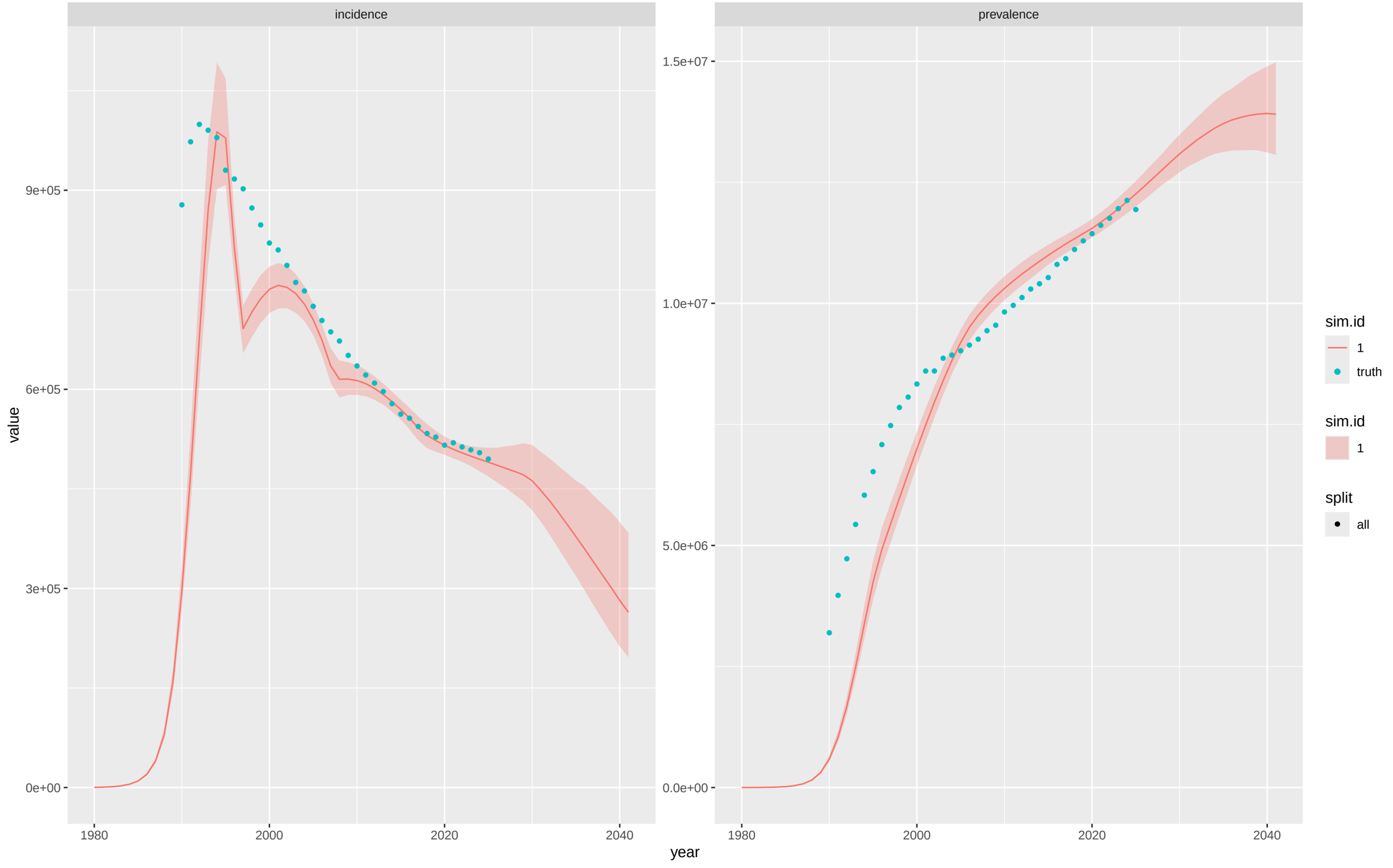
unaids.remainder step 2: hiv.mortality.by.age (one color per chain)



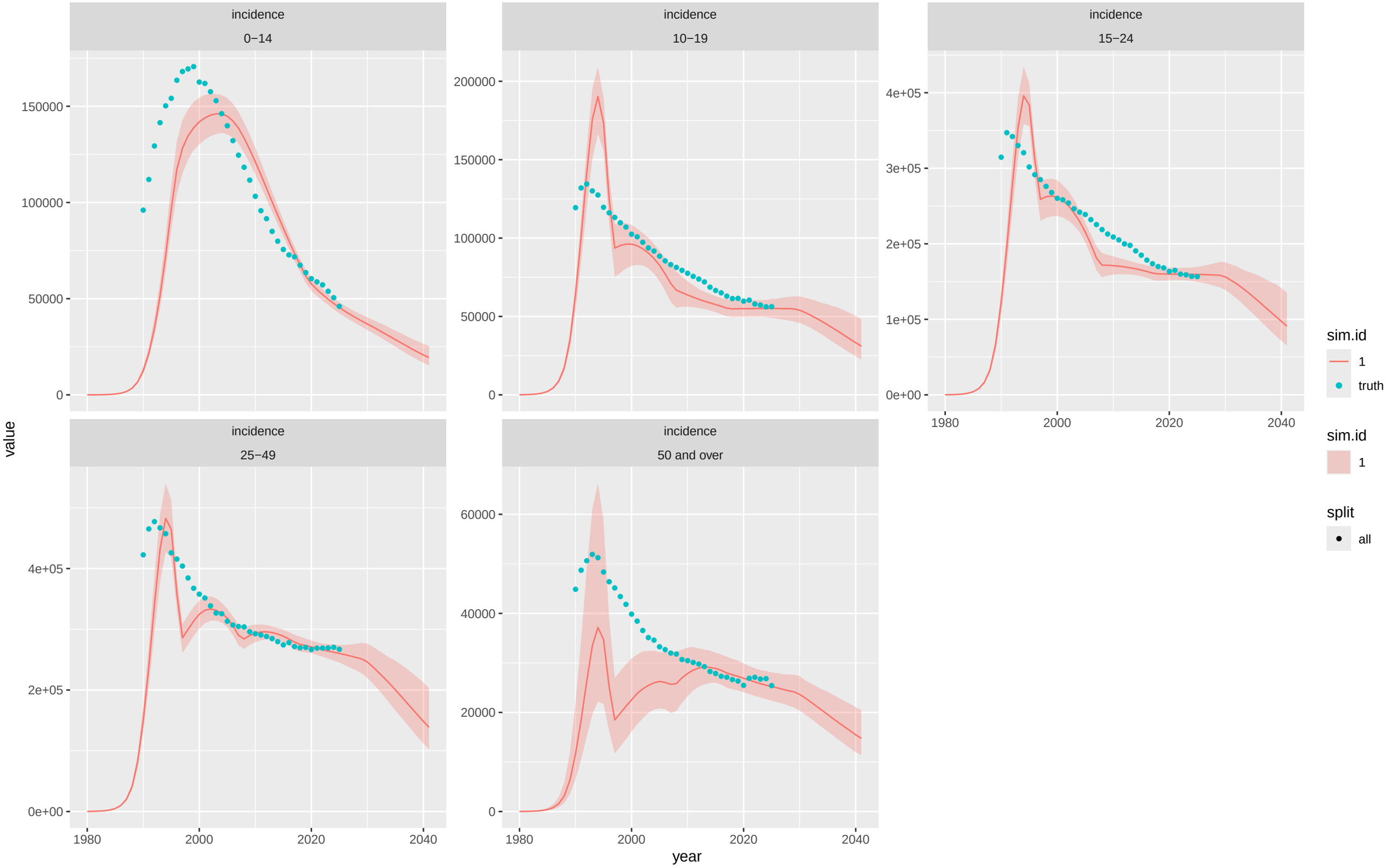
unaids.remainder step 2: hiv.mortality.by.age.sex (one color per chain)



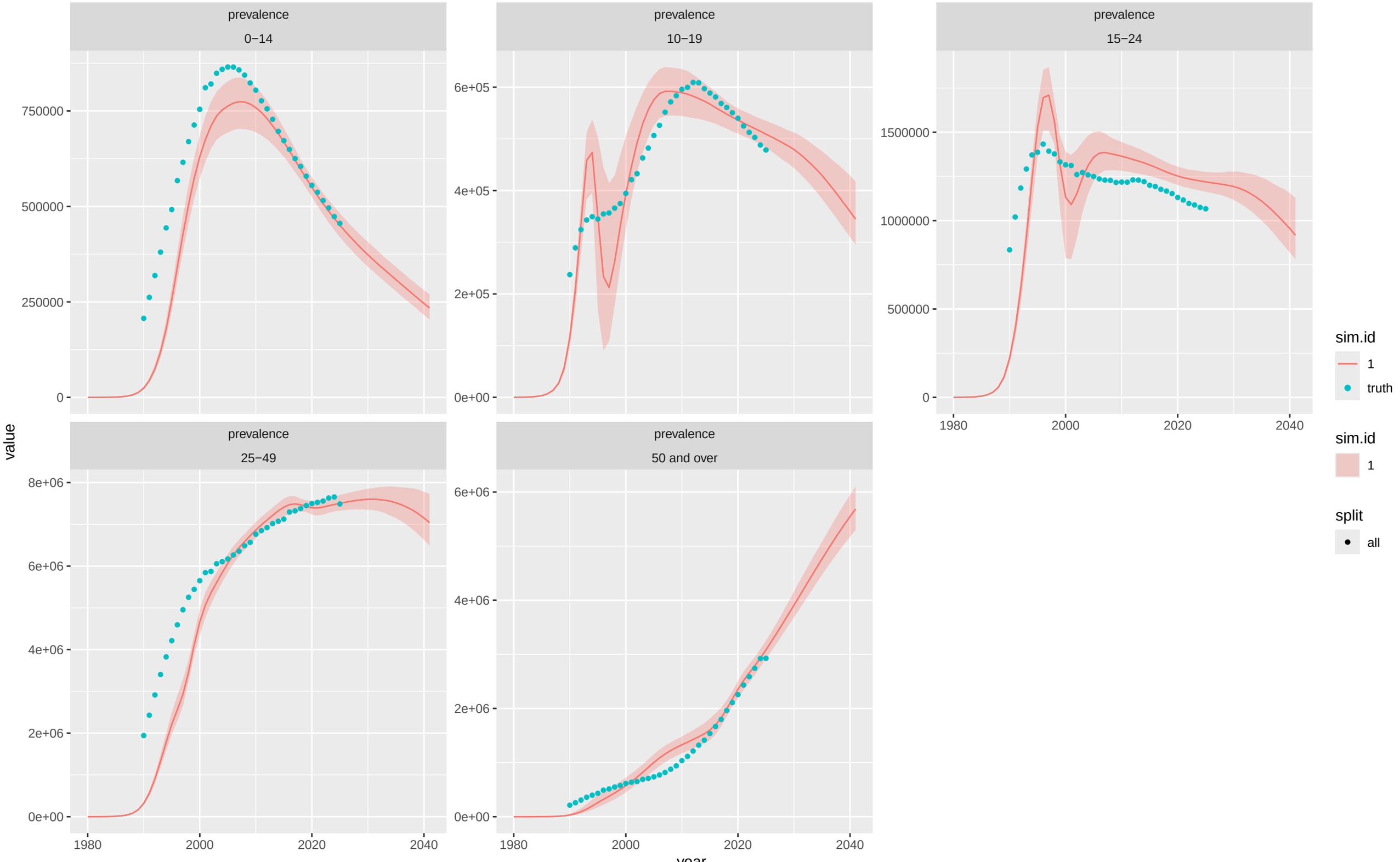
unaids.remainder: fit (merged Step 2 simset)



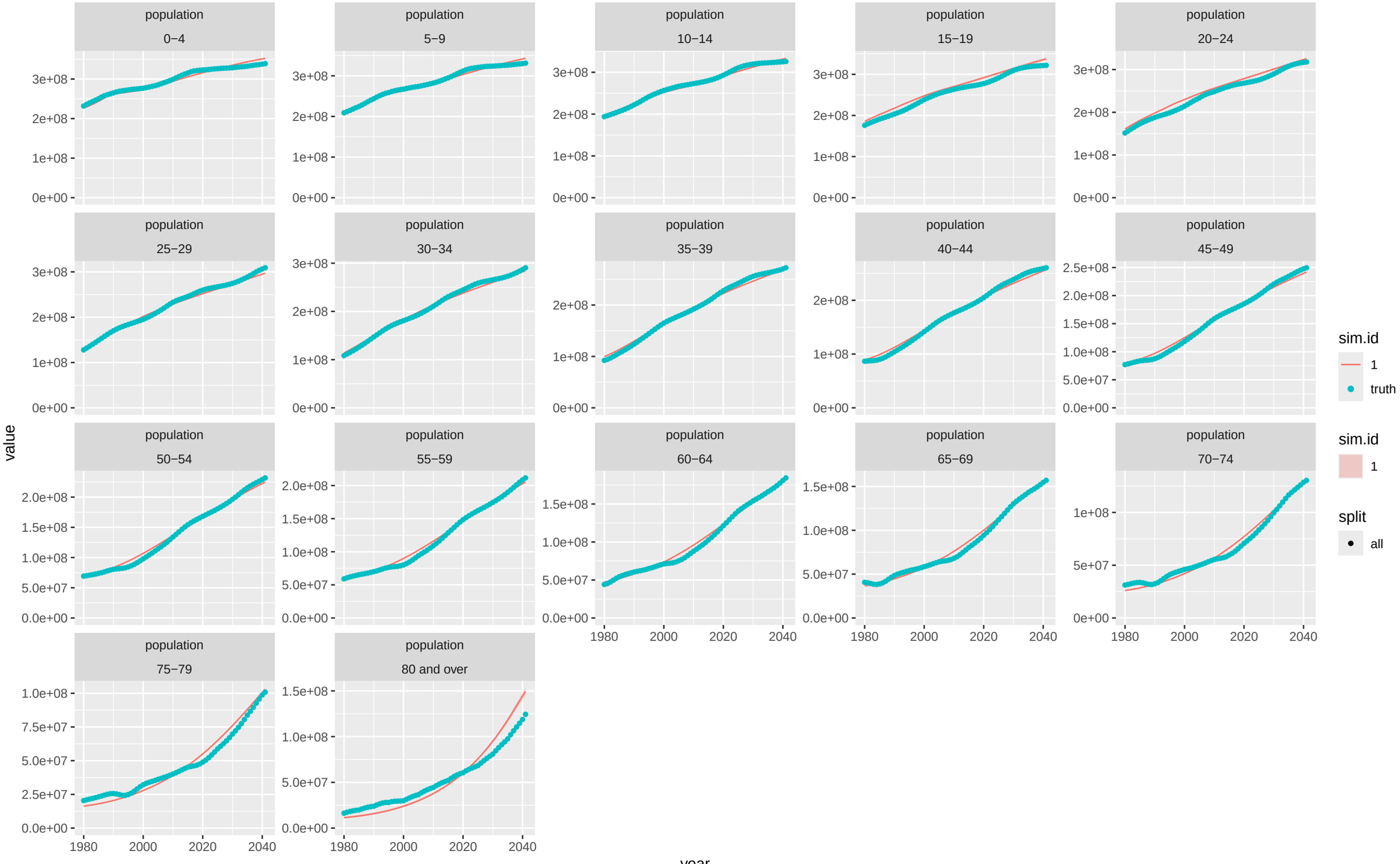
unaids.remainder: incidence.by.age (merged Step 2 simset)



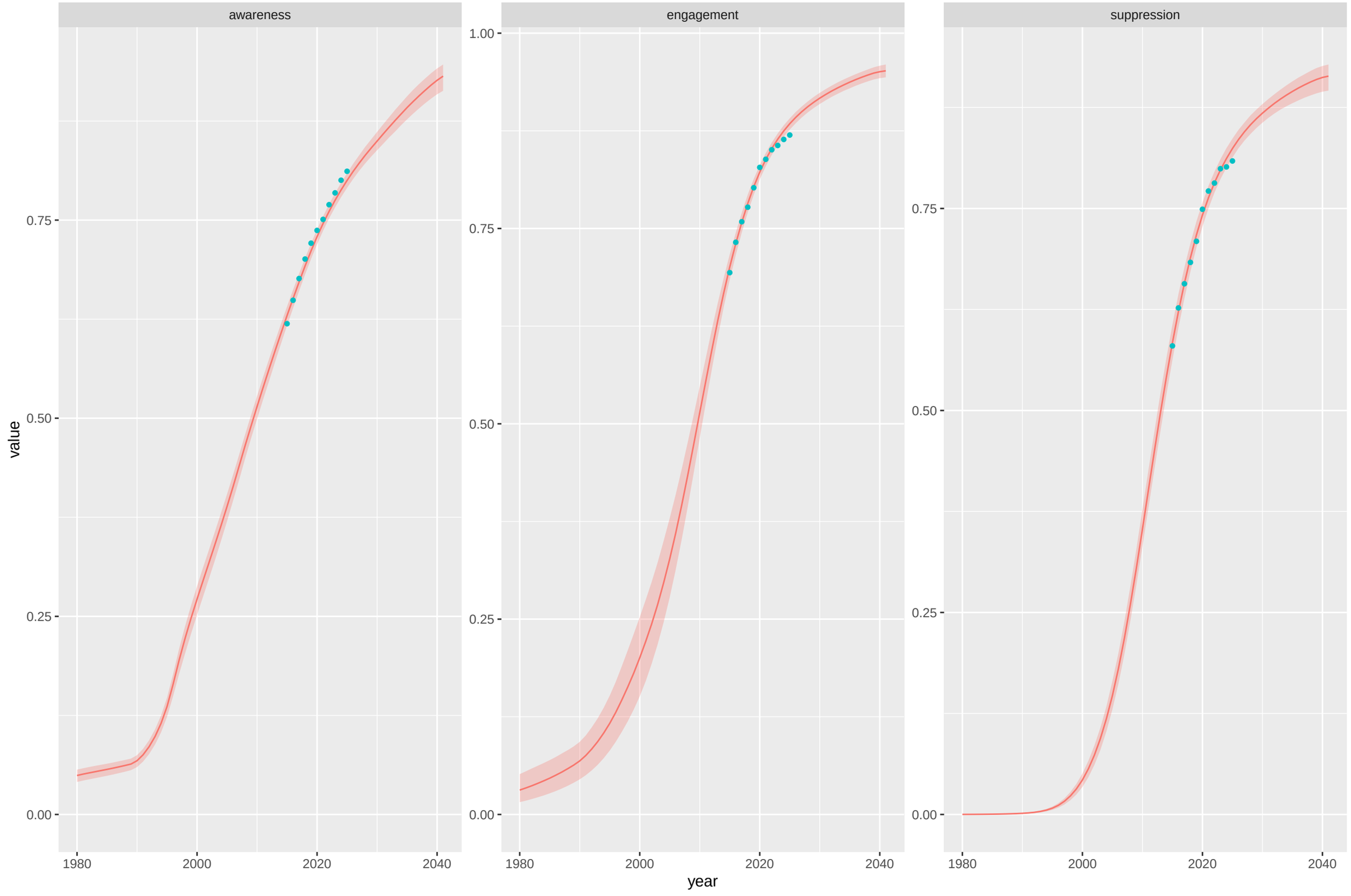
unaids.remainder: prevalence.by.age (merged Step 2 simset)



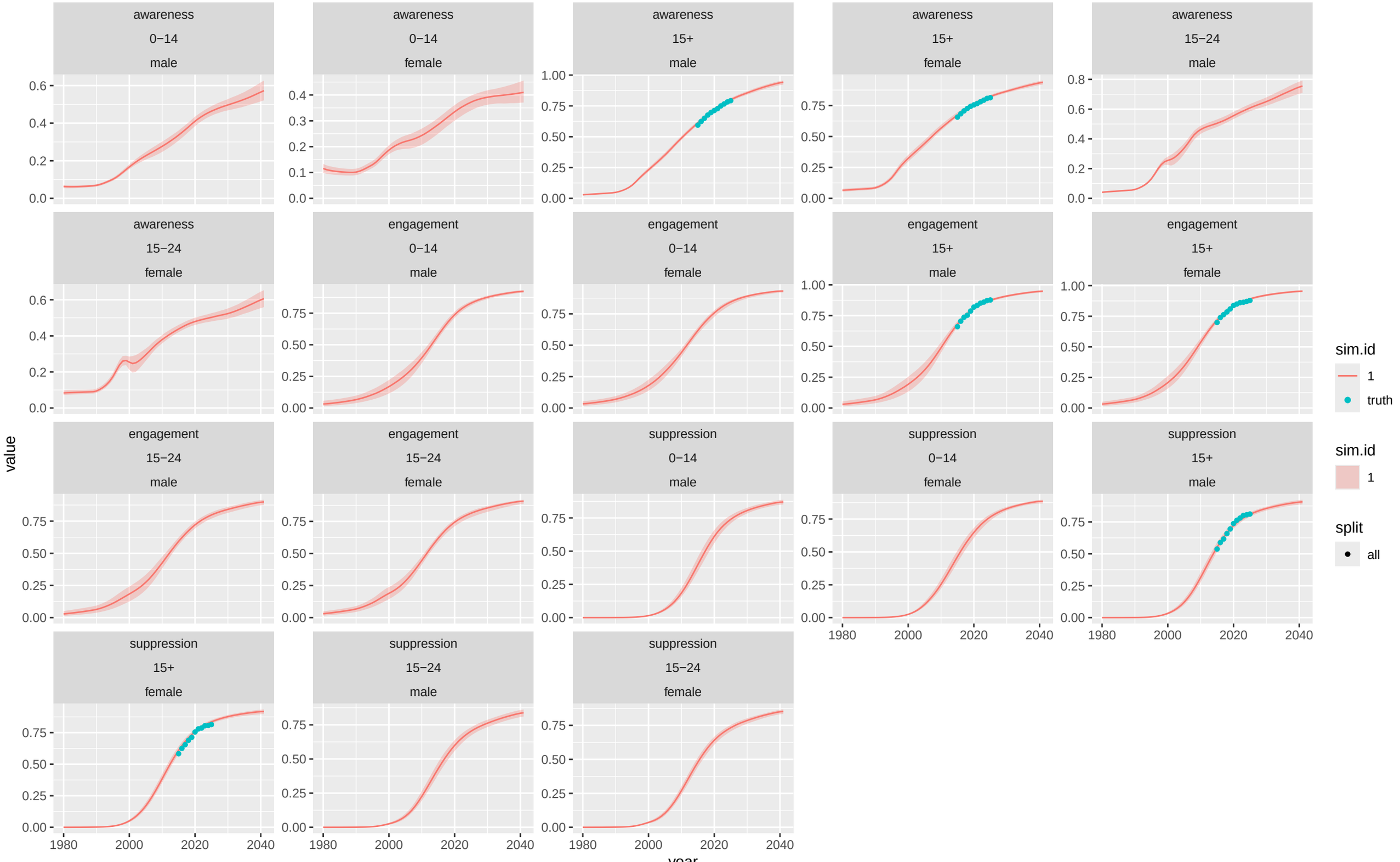
unaids.remainder: population.by.age (merged Step 2 simset)



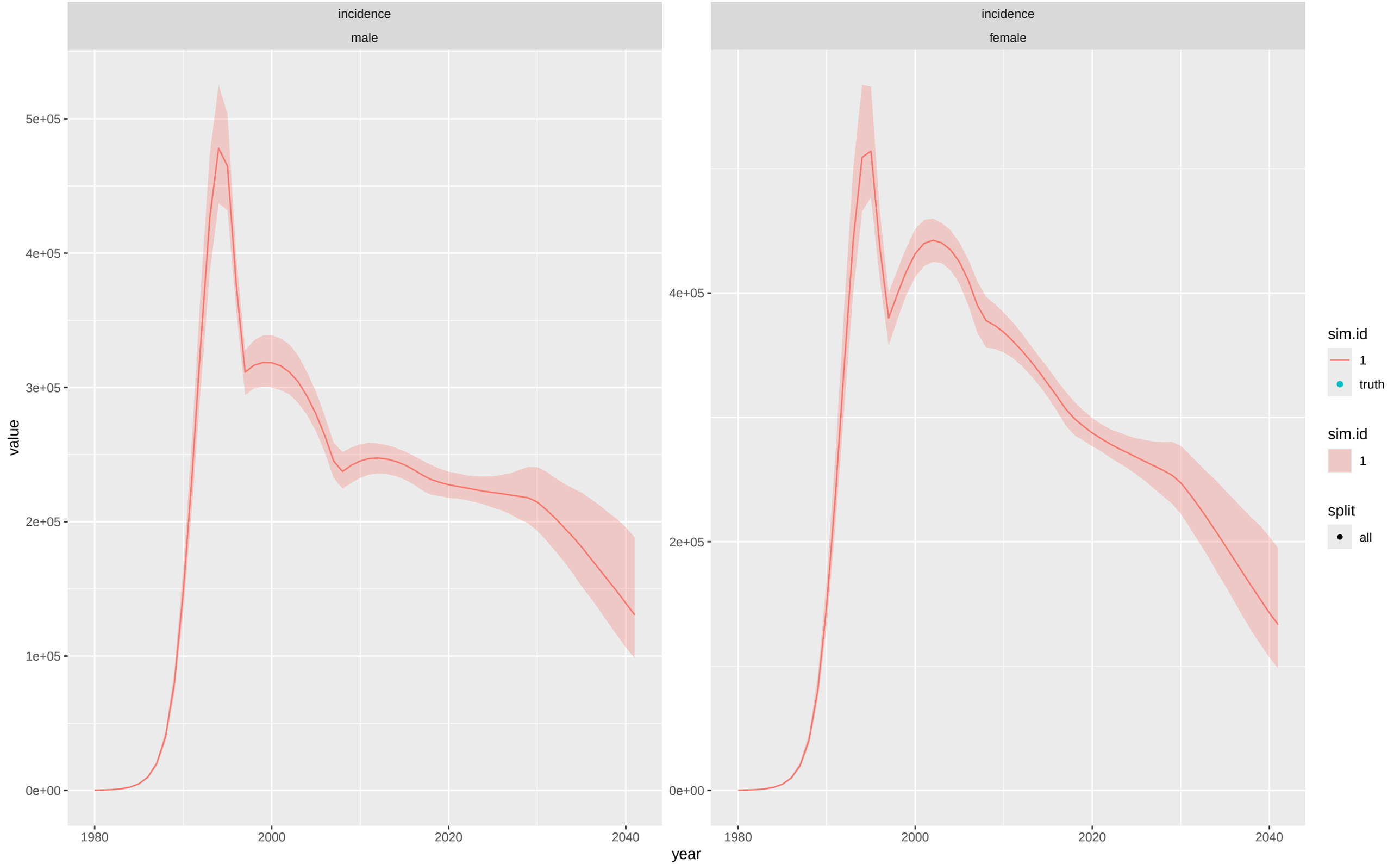
unaids.remainder: cascade (merged Step 2 simset)



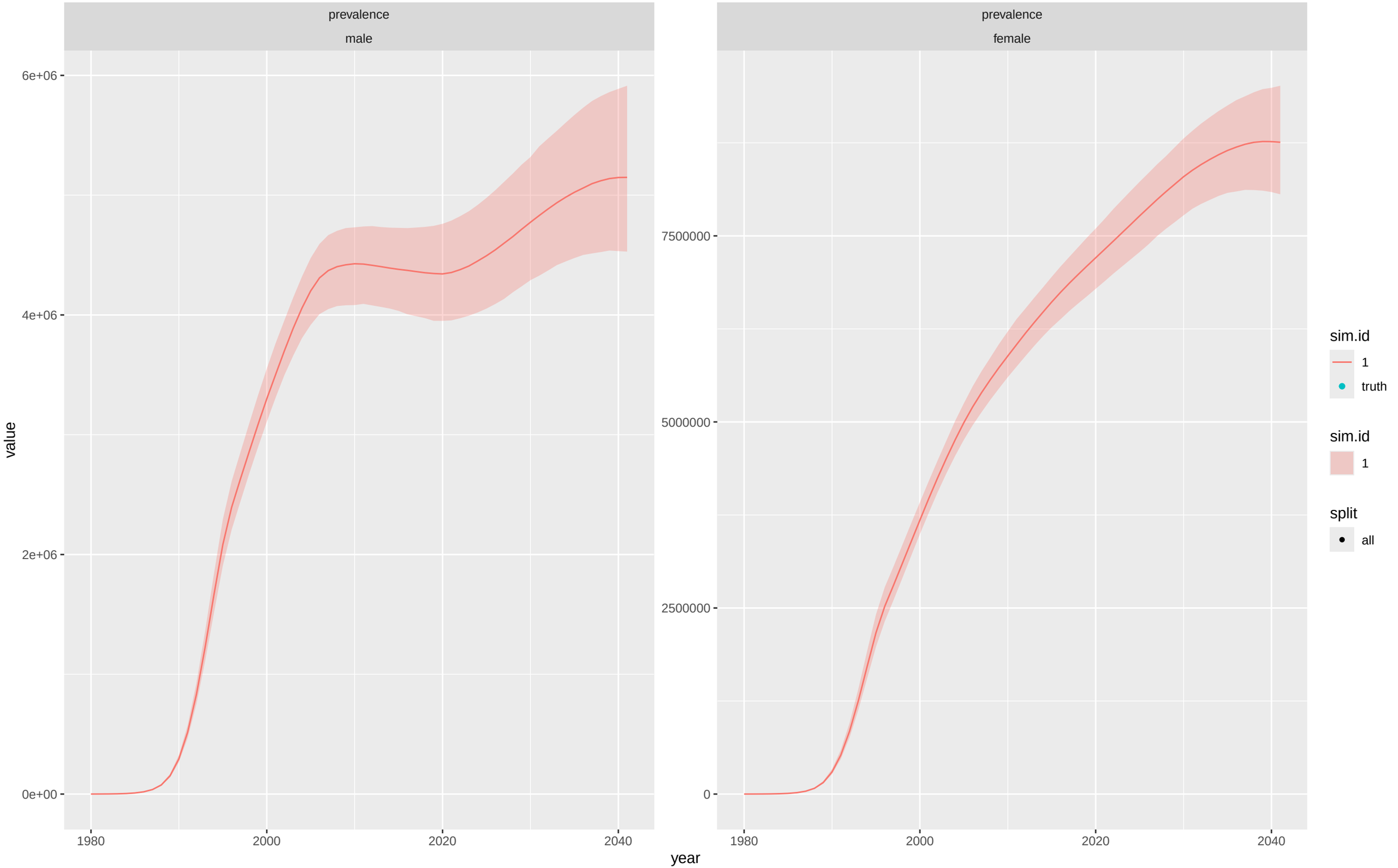
unaids.remainder: cascade.by.age.sex (merged Step 2 simset)



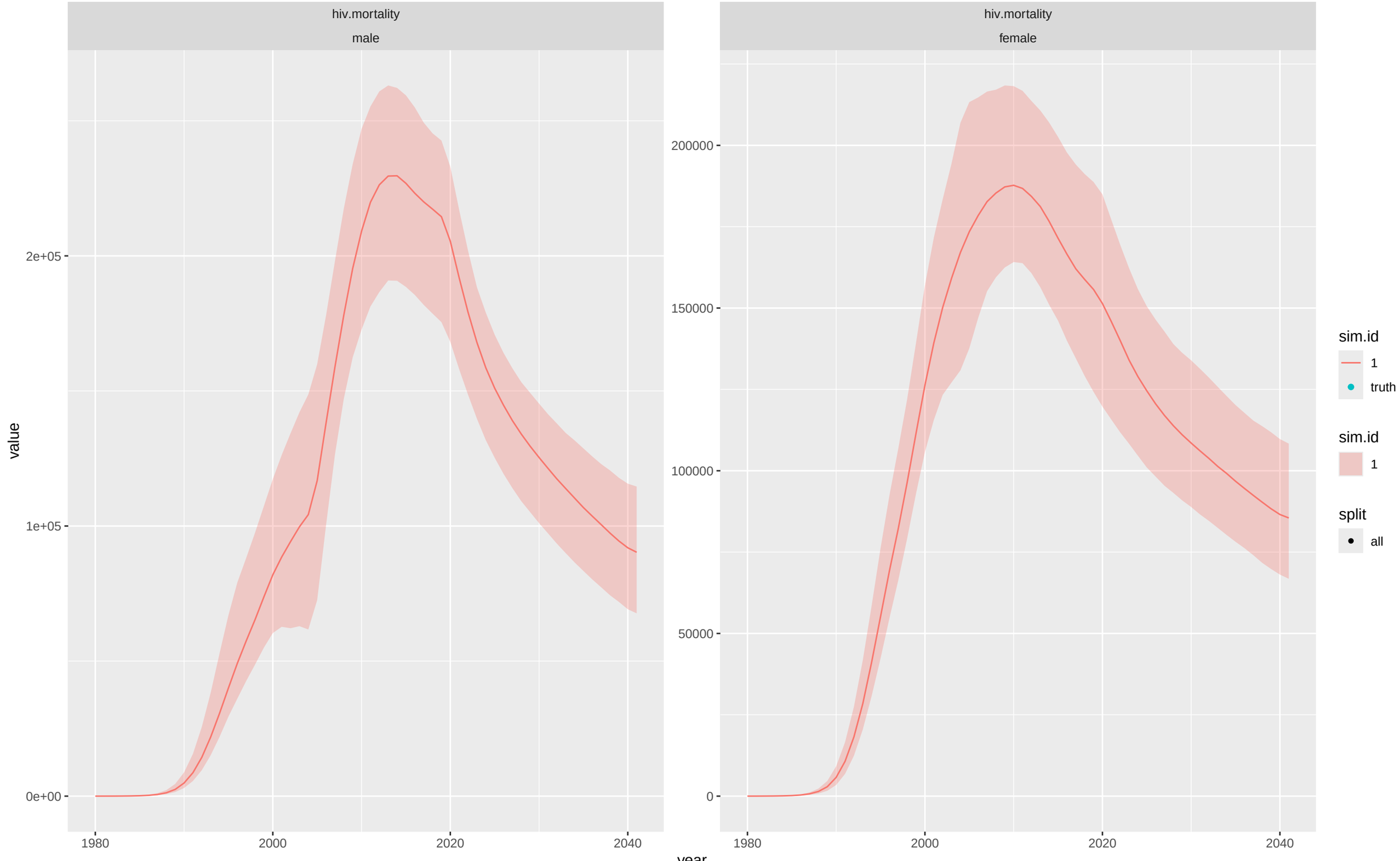
unaids.remainder: incidence.by.sex (merged Step 2 simset)



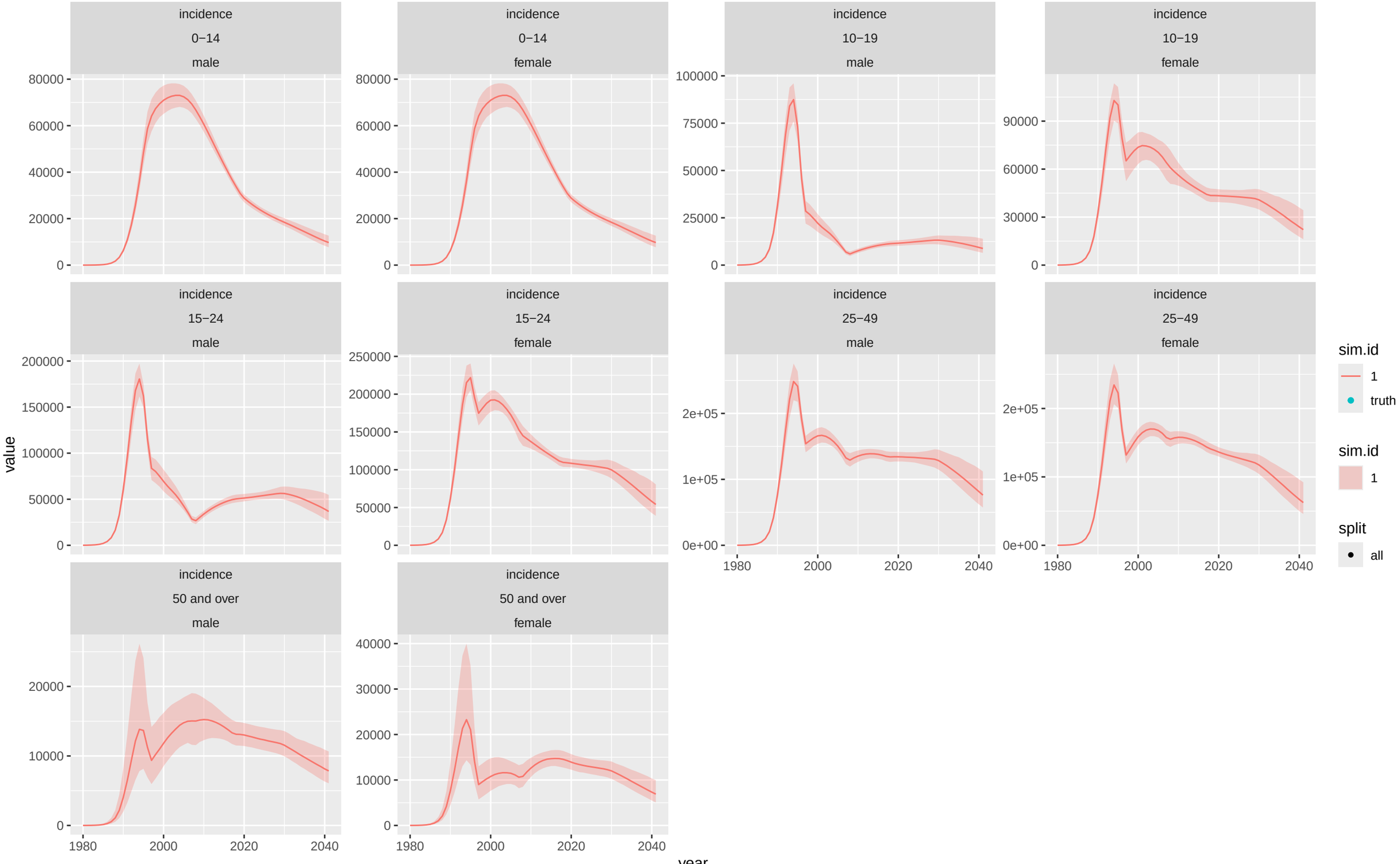
unaids.remainder: prevalence.by.sex (merged Step 2 simset)



unaids.remainder: hiv.mortality.by.sex (merged Step 2 simset)



unaids.remainder: incidence.by.age.sex (merged Step 2 simset)



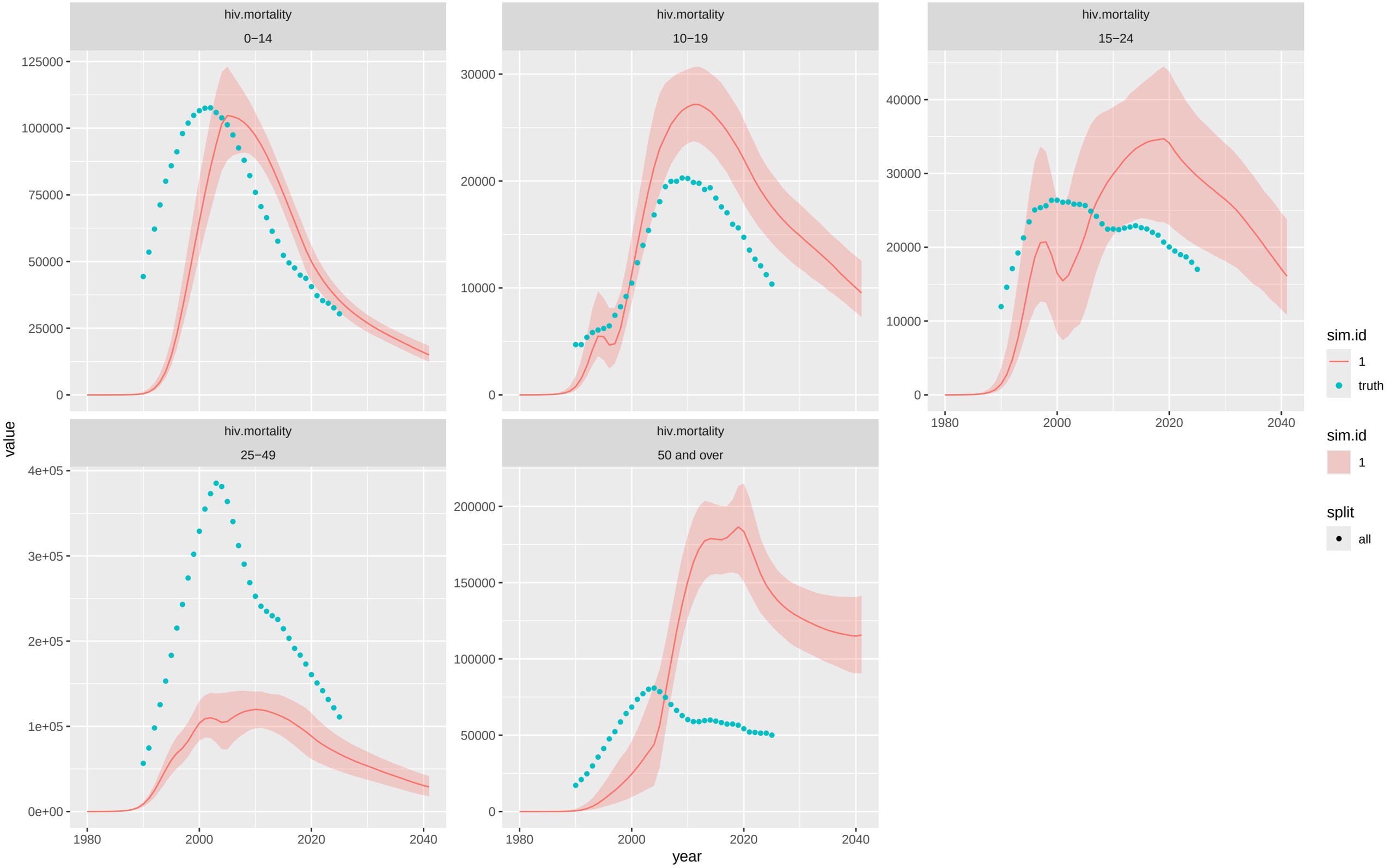
unaids.remainder: prevalence.by.age.sex (merged Step 2 simset)

prevalence 0–14 male
prevalence 0–14 female
prevalence 10–19 male
prevalence 10–19 female
prevalence 15–24 male
prevalence 15–24 female
prevalence 25–49 male
prevalence 25–49 female
prevalence 50 and over male
prevalence 50 and over female

value
year

sim.id
1
truth
split
all

unaids.remainder: hiv.mortality.by.age (merged Step 2 simset)



unaids.remainder: hiv.mortality.by.age.sex (merged Step 2 simset)

